

Spatial Mismatch, Segregation, Neighborhood Effects And Job Access

Special Topic 4: Urban and Labor – Week 3

Central question

- How do residential segregation and neighborhood exposure shape access to jobs, networks, and long-run labor-market opportunity?
- This is a labor question because place changes search, hiring, commuting, networks, skills, earnings, employment, and mobility.
- Week 3 discipline: separate mechanisms before interpreting neighborhood or segregation gaps.
- The goal is unequal access, not aggregate access.

From aggregate access to unequal access

- Week 1: local labor markets are systems of jobs, residences, commuting costs, rents, amenities, and outside options.
- Week 2: housing and moving costs can block access to high-opportunity places.
- Week 3: workers in the same metro area can face different feasible job sets.
- Strong metro labor demand can coexist with weak access for segregated or transit-constrained neighborhoods.

$$A_i = \sum_j v_j \exp\{-\kappa \tau_{ij}\}$$

Term	Interpretation
A_i	effective job access for worker or residence i
v_j	job opportunities, vacancies, or job quality at location j
τ_{ij}	generalized travel cost: time, money, reliability, transfers, safety, schedules
κ	rate at which access decays with travel cost

Mechanism taxonomy

Mechanism	Main labor object
Distance to jobs	reachable vacancies and commute feasibility
Transit and schedules	generalized travel cost and feasible shifts
Address stigma	employer callbacks, interviews, offers
Residential networks	referrals, information, coworker links
Peers and social capital	search behavior, aspirations, expectations
Schools and institutions	human capital and long-run opportunity
Safety and exposure	work timing, search radius, retention, welfare

Mechanism decomposition

$$Y_i = \beta_A A_i + \beta_N N_i + \beta_P P_i + \beta_S S_i + u_i$$

- A_i : job access and commuting feasibility.
- N_i : referral networks and neighbor-coworker links.
- P_i : peers, social capital, safety, local expectations.
- S_i : schools, public goods, and local institutions.
- The warning: one neighborhood coefficient can bundle several mechanisms.

Spatial mismatch, networks, neighborhood effects

Channel	Counterfactual	Natural design
Spatial mismatch	same worker, lower travel cost to jobs	travel-time, transit, job-location shocks
Networks	same worker, different referral environment	matched worker-firm and neighbor links
Address stigma	same applicant, different address signal	audit or correspondence experiment
Neighborhood exposure	same child, different duration or timing of exposure	mover or policy-induced relocation design

Job suburbanization and unequal access

- Jobs can move even when residential segregation persists.
- A suburban job center may raise total metro employment but lower effective access for transit-dependent central-city residents.
- The labor margin may be employment, commute burden, job quality, or search scope.
- Proximity is not access: employers, referrals, schedules, and transport all mediate the connection.

Neighborhood exposure and long-run outcomes

$$Y_i = \alpha + \theta \text{Exposure}_{ig} + \lambda_g + \varepsilon_i$$

- Exposure designs ask whether duration or age at exposure changes adult outcomes.
- Earlier moves can reveal causal place effects when timing is plausibly comparable.
- Labor outcomes include later earnings, employment, college, occupational mobility, and search scope.
- Mechanisms may run through schools, peers, safety, institutions, networks, or expectations.

Mover designs and policy shocks

Design	Best for	Main threat
Mover / age-at-move	long-run exposure timing	endogenous moves and disruption
Cohort exposure	cumulative duration and age profiles	omitted family trends
MTO or vouchers	neighborhood opportunity change	treatment bundles
Public housing demolition	policy-induced relocation	displacement channels
Transit or job-location shock	current access change	equilibrium resorting

School-boundary, audit, matched-firm designs

Design	Best use	Key limit
School boundary or admission	separate schools from neighborhoods	capitalization
Audit or correspondence	address stigma, distance, employer beliefs	callback margin
Matched employer-employee	referrals, coworker geography, workplace segregation	data access
Neighbor-coworker overlap	residential labor-market networks	sorting

What good empirical design does

- Maps the treatment to one main mechanism.
- Defines geography using commuting, schools, institutions, or employer markets.
- Measures actual access, exposure, or employer response.
- Distinguishes residence, workplace, school, and neighborhood channels.
- Addresses residential sorting and timing.
- Discusses resorting, displacement, and capitalization when relevant.
- States which labor margin moves.

Data sources

Source	Typical use
LEHD / LODES / QWI	residence-workplace flows and job geography
Census / ACS / Decennial	segregation, commuting, earnings, sorting
Opportunity Atlas / tax-linked data	childhood exposure and adult outcomes
School boundaries / assignment	school versus neighborhood channels
GTFS / travel-time matrices	generalized job access
Audit data	employer response to address or commute signals
Matched employer-employee data	referrals and workplace segregation
Housing or relocation admin data	policy-induced moves and exposure

- Reproduce: synthetic age-at-move exposure logic anchored in MTO.
- Diagnose: classify mechanisms and identify the counterfactual.
- Transfer: apply the same discipline to job suburbanization.
- Goal: separate access from employment, neighborhood effects from sorting, schools from neighborhoods, and policy shocks from mover designs.

- Week 3 treats safety as one mechanism inside unequal access.
- Week 4 studies safety and environmental exposure directly as constraints on work.
- The same design discipline carries forward: define the labor margin, mechanism, geography, and counterfactual.

- Segregation matters for labor when it changes access, networks, stigma, schools, exposure, safety, or institutions.
- Proximity to jobs is not the same as access to jobs.
- Neighborhood effects require designs that confront sorting and channel bundling.
- The best papers identify a mechanism and a labor margin, not a generic place effect.